Information Theory Workshop (ITW) 2026

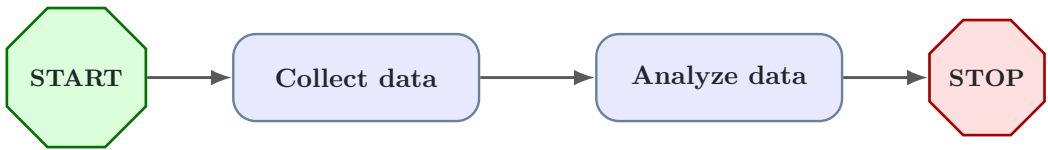
Information-theoretic foundations of anytime-valid inference

Part 1 (9:00 am to 10:30 am)

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November 10th, 2026

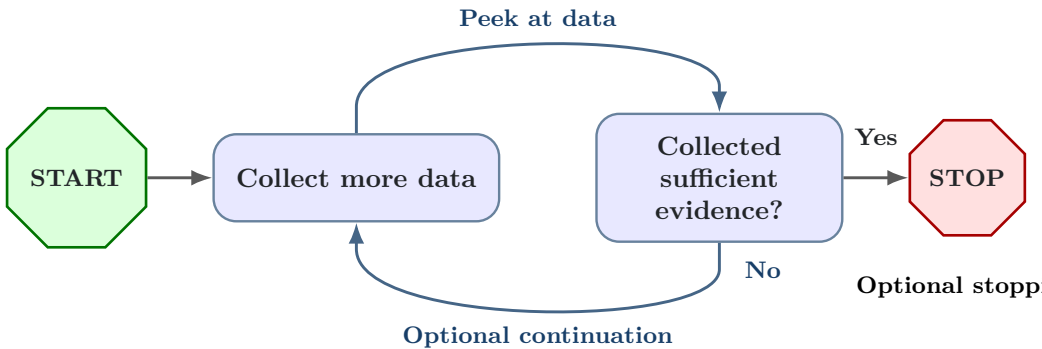


Statistical inference: extract information from data to draw reliable conclusions.



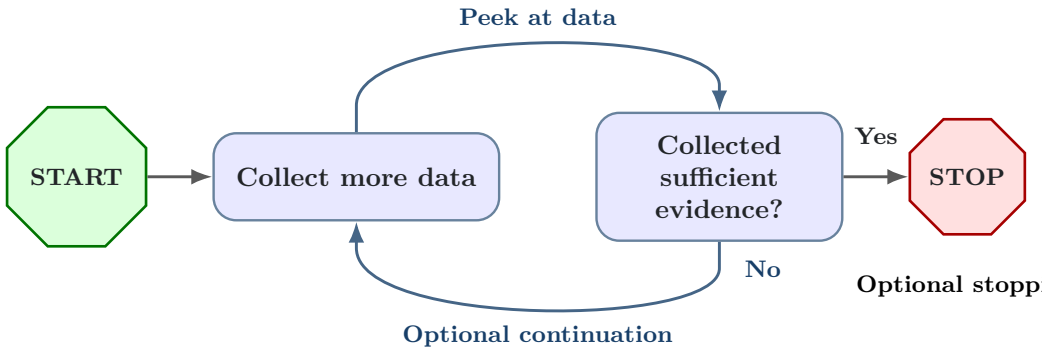
Statistical inference: extract information from data to draw reliable conclusions.

Anytime-valid inference: retain validity under *peeking*, *optional stopping* and *continuation*.



Statistical inference: extract information from data to draw reliable conclusions.

Anytime-valid inference: retain validity under *peeking*, *optional stopping* and *continuation*.



This tutorial: how information theoretic ideas can be used to understand the fundamental limits and identify general design principles for anytime-valid inference problems.

Tutorial outline

PART 1:
Introduction + finite alphabet methods

1. Motivation
A/B testing example; failure of classical methods; Anytime-valid (AV) testing

2. Fundamental Limits
Classical DPI + Wald arguments

3. AV Test for Finite Alphabets
Universal Inference; Proof of optimality

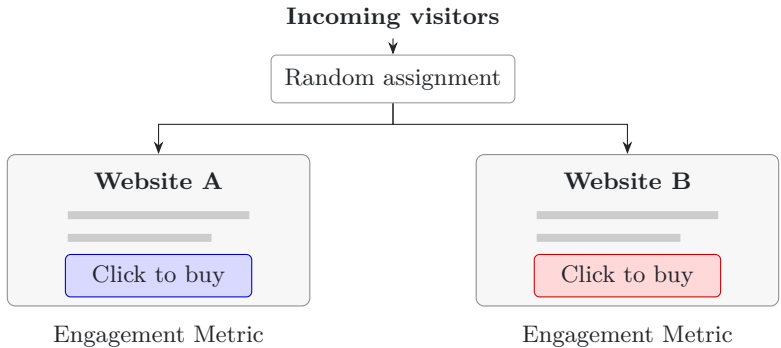
PART 2:
General Principles

1. Bounded mean testing
Testing-by-betting principle, Kelly criterion, universal portfolios.

2. Unifying Design Principle
Donsker-Varadhan representation, Likelihood-free Universal Inference.

3. Extensions
Sequential estimation, change detection, non-iid data, etc.

Motivating application: A/B testing



Goal: Reliably identify the better version using as little data as possible?

A simplified formulation

Suppose we pair visitors off, assigning one to each website.

- Let $U_i, V_i \in \mathbb{R}$ denote the outcomes for the i-th pair (e.g., time spent on the website).
- Record whether A outperformed B:

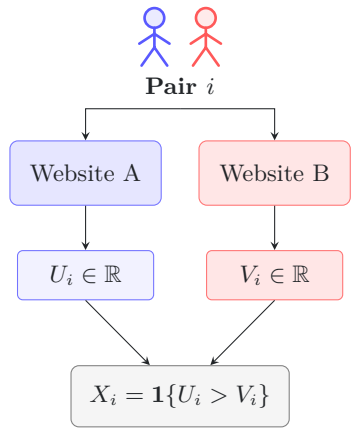
$$X_i = \mathbf{1}\{U_i > V_i\}.$$

- Assume i.i.d. pairs, with no ties, and define

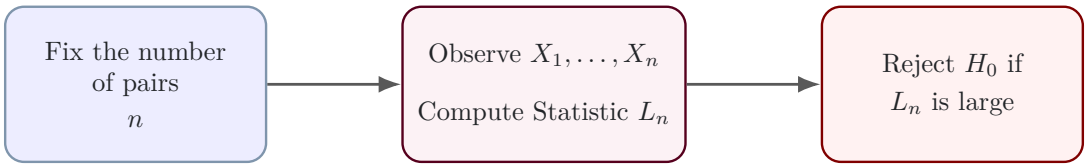
$$p = \mathbb{P}(U_i > V_i).$$

Our testing problem:

$$H_0 : p \leq \tfrac{1}{2} \quad \text{vs.} \quad H_1 : p > \tfrac{1}{2}$$



Method 1: Fixed-sample testing



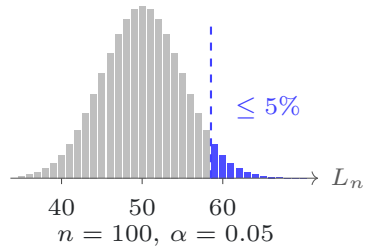
Method 1: Fixed-sample testing

- Choose the sample size n **before** observing the data
- With $S_n = \sum_{i=1}^{n} X_i$, define the generalized likelihood-ratio statistic

$$L_n = \frac{\sup_{q \in [1/2,1]} q^{S_n}(1-q)^{n-S_n}}{(1/2)^n}$$

- L_n is expected to be small under H_0
- Reject H_0 when $L_n \geq B_{\alpha,n}$, where $B_{\alpha,n}$ ensures

$$\sup_{p \leq 1/2} \mathbb{P}_p(L_n \geq B_{\alpha,n}) = \mathbb{P}_{1/2}(L_n \geq B_{\alpha,n}) \leq \alpha$$



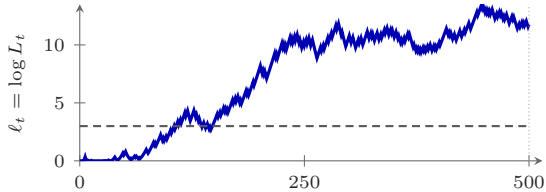
The guarantee applies only at the preselected sample size n

Some limitations of fixed-sample testing

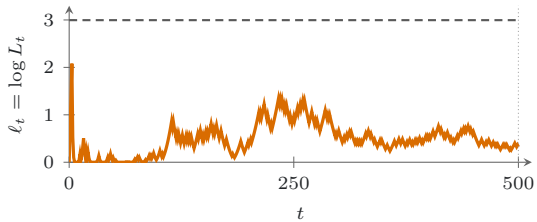
- Fix the sample size at $n = 500$ **pairs**.
- Consider two cases:
  1. **Strong effect** ($p = 0.60$)
  2. **Weak effect** ($p = 0.52$)
- For $p = 0.60$, it appears that we could have stopped at $t \ll 500$
- For $p = 0.52$, the evidence at $n = 500$ is insufficient

Committing to a fixed sample-size n can either waste resources or lead to insufficient evidence.

Strong effect: $p = 0.60$

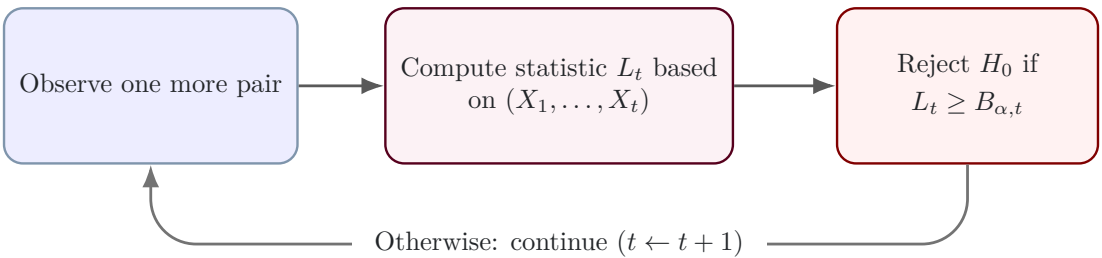


Weak effect: $p = 0.52$



Method 2: Monitor the stream continuously

Change: Run Method 1 repeatedly as new observations arrive.



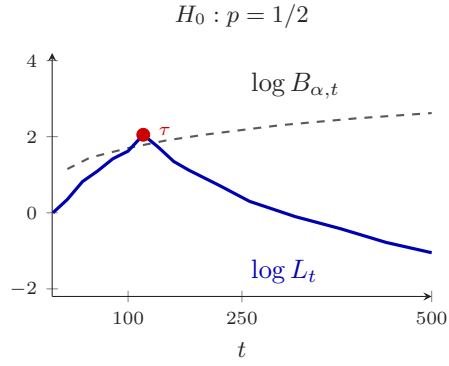
Method 2: Monitor the stream continuously

Idea: Repeatedly apply the fixed-sample test at each time $t = 1, 2, \ldots$

Stop at the first time t at which the fixed-sample test rejects H_0:

$$\tau \;=\; \inf\{t \geq 1 : \log L_t \geq \log B_{\alpha,t}\}.$$

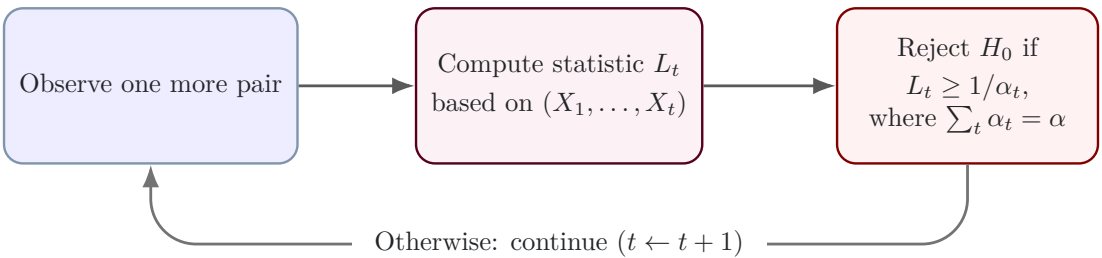
- This creates **many opportunities** for a false rejection under H_0
- It leads to an inflation of false-alarm rate
- This phenomenon is known as **sampling to a foregone conclusion**.



A False Alarm: The test rejects H_0 at $\tau = 120$

Method 3: Bonferroni correction

Change: Allocate an error budget to each "peek" at the data.



Method 3: Bonferroni correction

- Allocate an error budget to each "peek" at the data:

$$\alpha_t = \frac{6\alpha}{\pi^2 t^2}, \qquad \sum_{t \geq 1} \alpha_t = \alpha.$$

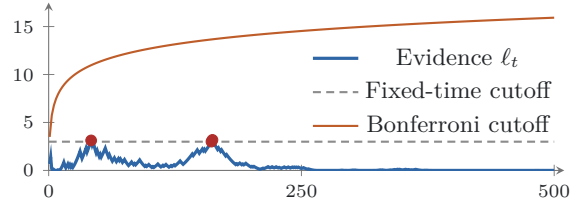
- Reject H_0 when

$$\ell_t = \log L_t \geq \log(1/\alpha_t).$$

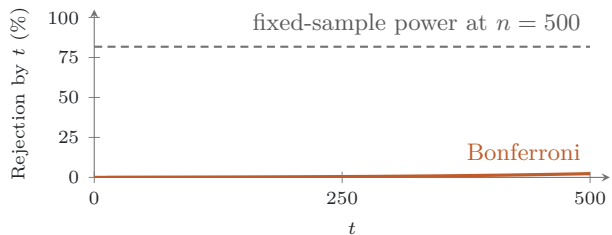
- The **union bound** guarantees

$$\sup_{p \leq 1/2} \mathbb{P}_p(\exists t : \ \ell_t \geq \log(1/\alpha_t)) \leq \alpha.$$

Under $H_0 : p = 1/2$

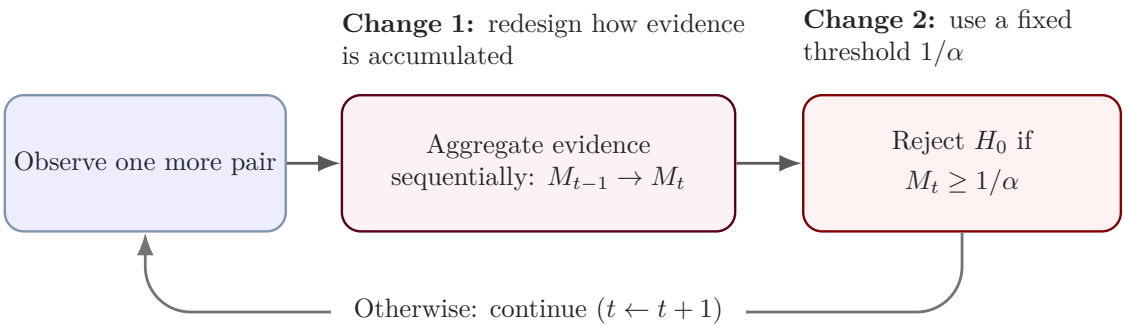


Under $H_1 : p = 0.575$



Method 4: Anytime-valid testing

Anytime-valid testing changes the design principle: rather than repeatedly reusing a fixed-time test, we construct a new evidence process designed for continuous monitoring.



Method 4: Anytime-valid testing

Anytime-valid procedures offer the right balance between naive peeking and conservative Bonferroni corrections.

The central idea is to construct a nonnegative process $\{M_t : t \geq 0\}$, using only the data observed so far, such that

1. Under H_0, the process is unlikely to take large values.
2. Under H_1, the process grows rapidly, allowing us to stop early.

We then reject H_0 at the first threshold crossing:

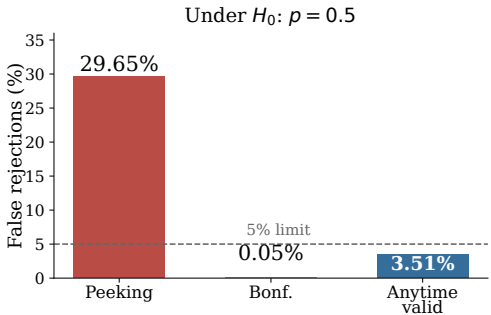
$$\tau_\alpha = \inf\{t \geq 1 : M_t \geq 1/\alpha\}.$$

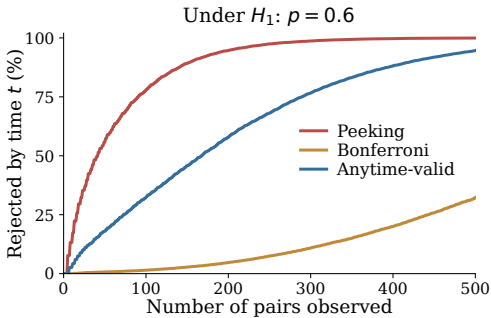
Method 4: Anytime-valid testing

For the A/B testing example, an appropriate choice of $\{M_t : t \geq 1\}$ is

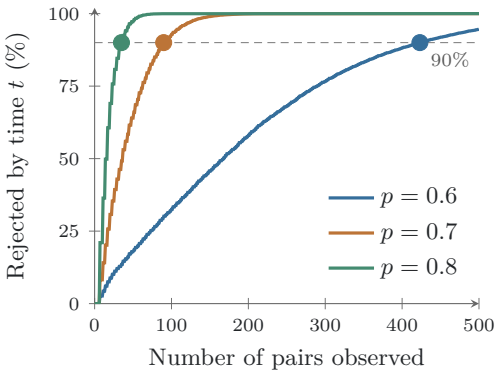
$$M_t = \frac{\prod_{i=1}^{t} \widehat{p}_i^{X_i}(1-\widehat{p}_i)^{1-X_i}}{(1/2)^t}, \qquad \widehat{p}_i = \frac{1/2 + \sum_{j=1}^{i-1} X_j}{i}$$

The stopping rule $\tau_\alpha = \inf\{t \geq 1 : M_{\geq} 1/\alpha\}$ satisfies $\sup_{p \leq 1/2} \mathbb{P}_p(\tau < \infty) \leq \alpha$

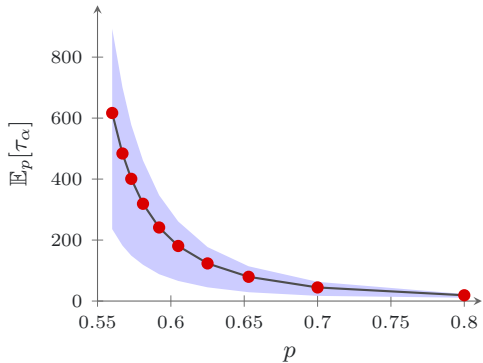




Anytime-valid tests adapt to the effect size



Stronger signal ⇒ earlier stopping.



Mean (red) and middle 50% (shaded).

Summary and looking ahead

Fixed-sample testing	**Naive peeking**	**Bonferroni**	**Anytime-valid**
Valid at a fixed sample size	Repeated looks inflate false alarms	Valid across time, but conservative	Valid across time **and adaptive**

Rest of the tutorial

1. **Fundamental limits:** How quickly can *any* valid test stop?
2. **General design principles:** How do we construct the "evidence process" $\{M_t : t \geq 1\}$?
3. **Optimality:** When is a given test the best possible?

Formal Definitions and Fundamental Limits

Anytime-valid testing

- An i.i.d. stream of $\mathcal{X}$ valued observations

$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P_X, \qquad P \in \mathcal{P}(\mathcal{X})$$

- Filtration $\{\mathcal{F}_t : t \geq 1\}$ with $\mathcal{F}_t \coloneqq \sigma(X_1, \ldots, X_t)$

- Given two disjoint $\mathcal{P}_0, \mathcal{P}_1 \subset \mathcal{P}(\mathcal{X})$, decide between

$$H_0 : P_X \in \mathcal{P}_0, \quad \text{versus} \quad H_1 : P_X \in \mathcal{P}_1$$

- **Goal:** Define a (possibly infinite) $\{\mathcal{F}_t\}$-stopping time τ satisfying

$$\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau < \infty) \leq \alpha \qquad \text{“level-}\alpha\text{” property}$$

- τ denotes the time to stop collecting observations and reject H_0

Power and efficiency under the alternative

- **Power one.** We want the test to eventually reject under every alternative:

$$\mathbb{P}_P(\tau_\alpha < \infty) = 1, \qquad \forall P \in \mathcal{P}_1.$$

  Often, we take $\mathcal{P}_1 = \mathcal{P}(\mathcal{X}) \setminus \mathcal{P}_0$.

- **Efficiency.** Let $\mathcal{T}_\alpha \equiv \mathcal{T}_\alpha(\mathcal{P}_0)$ denote the collection of all level-α tests:

$$\mathcal{T}_\alpha = \left\{ \tau : \sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau < \infty) \leq \alpha \right\}.$$

  Among power-one tests in $\mathcal{T}_\alpha$, we seek to minimize $\mathbb{E}_P[\tau]$.

- **Focus of this tutorial.** For a fixed alternative $P \in \mathcal{P}_1$, characterize the term

$$\inf_{\tau_\alpha \in \mathcal{T}_\alpha(\mathcal{P}_0)} \mathbb{E}_P[\tau_\alpha] \qquad \text{as } \alpha \downarrow 0.$$

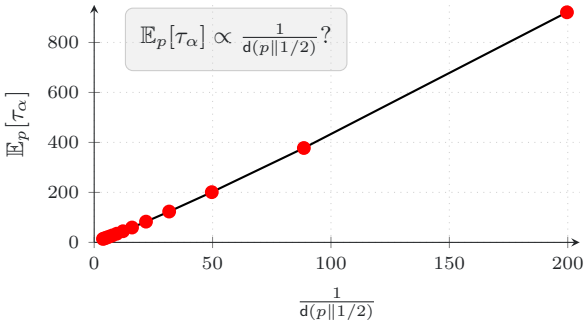
A/B testing example revisited

- $\mathcal{X} = \{0, 1\}$, and $P = \text{Bernoulli}(p)$
- $\mathcal{P}_0 = \{\text{Bernoulli}(p) : p \leq 1/2\}$
- $\mathcal{P}_1 = \{\text{Bernoulli}(p) : p > 1/2\}$
- We defined an AV test

$$\tau_\alpha = \inf\{t \geq 1 : M_t \geq 1/\alpha\}$$

where

$$M_t = \frac{\prod_{i=1}^t \widehat{p}_i^{Z_i}(1-\widehat{p}_i)^{1-Z_i}}{(1/2)^t}, \qquad \widehat{p}_i = \frac{1/2 + \sum_{j=1}^{i-1} X_j}{i}$$



A general lower bound

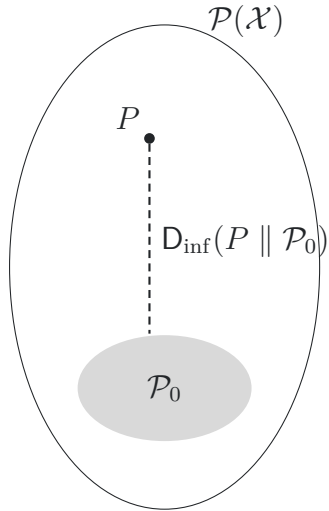
- Consider a general testing problem with $\mathcal{X}$-valued i.i.d. observations.
- Let $\mathcal{P}_0$ denote the null distribution class.
- Let $\mathcal{T}_\alpha(\mathcal{P}_0)$ denote the set of all level-α tests for the null $\mathcal{P}_0$.

Fundamental lower bound

Every $\tau \in \mathcal{T}_\alpha(\mathcal{P}_0)$ satisfies, for any $P \notin \mathcal{P}_0$,

$$\mathbb{E}_P[\tau] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)},$$

where $\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) = \inf_{Q \in \mathcal{P}_0} \mathsf{D}(P \parallel Q)$



Proof: Information in a stopped experiment

Fix $P \in \mathcal{P}_1$, $Q \in \mathcal{P}_0$ with $0 < \mathsf{D}(P \parallel Q) < \infty$.

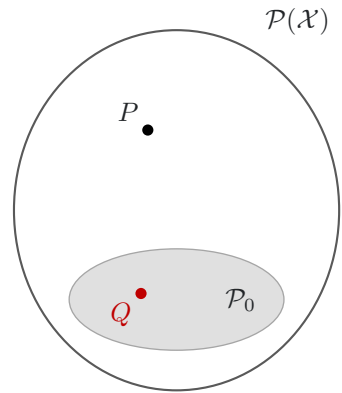
Assume $\mathbb{E}_P[\tau] < \infty$.

Stopped observations: $Y = (X_1, \ldots, X_\tau; \tau)$, with

$$\mathsf{P}_Y \coloneqq \mathrm{Law}_P(Y), \qquad \mathsf{Q}_Y \coloneqq \mathrm{Law}_Q(Y).$$

Relative entropy of the stopped experiment:

$$\begin{aligned}\mathsf{D}(P_Y \parallel Q_Y) &= \mathbb{E}_P\left[\log \frac{dP_Y}{dQ_Y}(Y)\right] \\ &= \mathbb{E}_P\left[\sum_{i=1}^{\tau} \log \frac{dP}{dQ}(X_i)\right].\end{aligned}$$



Proof: Apply Wald's identity

$$\mathsf{D}(\mathsf{P}_Y \parallel \mathsf{Q}_Y) = \mathbb{E}_P \left[\sum_{i=1}^{\tau} \log \frac{dP}{dQ}(X_i) \right].$$

Define the log-likelihood increments $Z_i := \log \frac{dP}{dQ}(X_i)$.

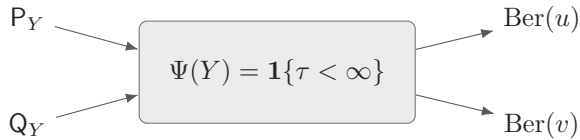
Conditions for Wald's identity:

- $\{Z_i\}_{i \geq 1}$ are i.i.d. under P, with $Z_i \in L^1(P)$ since $\mathsf{D}(P \parallel Q) < \infty$.
- τ is an $\{\mathcal{F}_t\}$-stopping time, and $\mathbb{E}_P[\tau] < \infty$.

$$\boxed{\mathsf{D}(\mathsf{P}_Y \parallel \mathsf{Q}_Y) = \mathbb{E}_P[\tau]\, \mathbb{E}_P[Z_1] = \mathbb{E}_P[\tau]\, \mathsf{D}(P \parallel Q).}$$

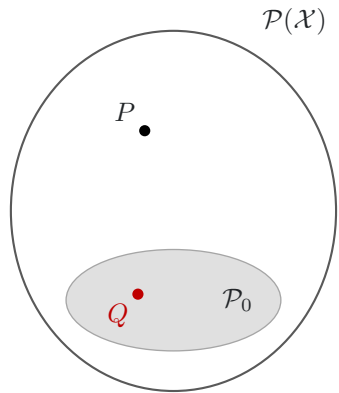
Proof: data processing for the decision

The test defines a **deterministic Markov kernel**:
$\Psi(Y) = \mathbf{1}\{\tau < \infty\}$.



$$u = \mathbb{P}_P(\tau < \infty) = 1, \qquad v = \mathbb{P}_Q(\tau < \infty) \le \alpha.$$

$$\mathbb{E}_P[\tau]\,\mathsf{D}(P \parallel Q) \overset{\text{Wald's}}{=} \mathsf{D}(P_Y \parallel Q_Y) \overset{\text{DPI}}{\ge} \mathsf{d}(1, v) \ge \log\frac{1}{\alpha}.$$



Taking an infimum over $Q \in \mathcal{P}_0$, we get $\mathbb{E}_P[\tau] \ge \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P\|\mathcal{P}_0)}$.

Summary so far

1. Problem. Given an i.i.d. stream of $\mathcal{X}$-valued observations, decide between

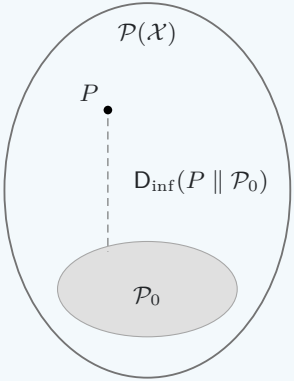
$$H_0 : P_X \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P_X \in \mathcal{P}_1.$$

2. Goal. Construct a level-α, power-one test τ_α satisfying

$$\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha < \infty) \le \alpha, \qquad \inf_{P \in \mathcal{P}_1} \mathbb{P}_P(\tau_\alpha < \infty) = 1.$$

3. Fundamental limit. Every such test satisfies

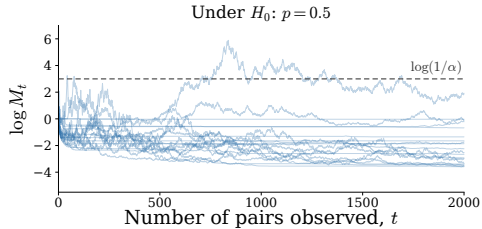
$$\mathbb{E}_P[\tau_\alpha] \ge \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}, \qquad P \in \mathcal{P}_1.$$



$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) = \inf_{Q \in \mathcal{P}_0} \mathsf{D}(P \| Q)$

How to construct anytime-valid tests?

- Recall that our general approach is to:
  - Construct a $\{\mathcal{F}_t\}$-adapted process $\{M_t : t \geq 1\}$ that quantifies the evidence against H_0
  - Define τ as the first time t such that M_t exceeds a threshold $1/\alpha$
- Requirements on the "evidence process" $\{M_t\}$:
  - Unlikely to take large values under H_0
  - Grows rapidly under H_1
- It can be shown that this is without loss of generality



The reverse information projection (RIPr)

Assume $\mathcal{P}_0$ is convex and compact.

The **RIPr** of P onto $\mathcal{P}_0$ is

$$Q^\star \equiv Q^\star(P, \mathcal{P}_0) \in \arg\min_{Q \in \mathcal{P}_0} \mathsf{D}(P\|Q).$$

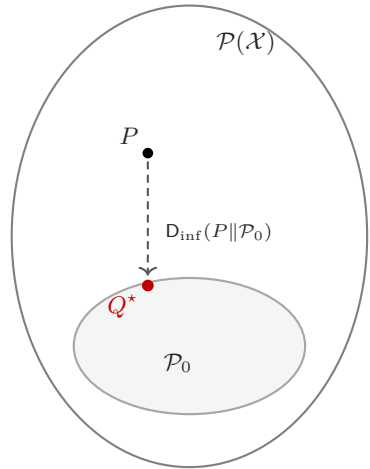
The crucial property: for every $Q \in \mathcal{P}_0$,

$$\mathbb{E}_Q\left[\frac{dP}{dQ^\star}(X)\right] \leq 1.$$

Interpretation: $dP/dQ^\star$ is unlikely to take large values under any $Q \in \mathcal{P}_0$.

A/B testing. For $P = \mathrm{Ber}(p)$ with $p > 1/2$ and null class $\mathcal{P}_0 = \{\mathrm{Ber}(q) : q \leq 1/2\}$,

$$Q^\star = \mathrm{Ber}(1/2).$$



The oracle evidence process

Construction. Suppose the alternative P is known. Using its reverse information projection $Q^\star$, define

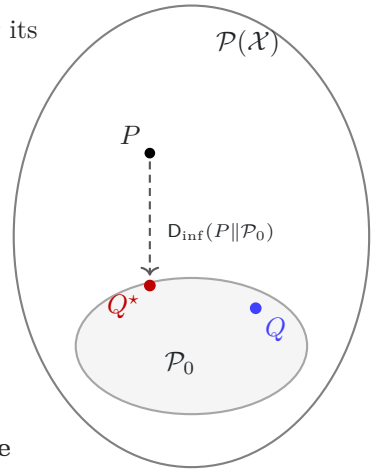
$$L_t^\star = \prod_{i=1}^{t} \frac{dP}{dQ^\star}(X_i), \qquad L_0^\star = 1.$$

Key observation. For every $Q \in \mathcal{P}_0$,

$$\mathbb{E}_Q[L_t^\star \mid \mathcal{F}_{t-1}] = L_{t-1}^\star \underbrace{\mathbb{E}_Q\left[\frac{dP}{dQ^\star}(X_t)\right]}_{\leq 1 \text{ (RIPr property)}}$$

$$\leq L_{t-1}^\star.$$

Conclusion. $\{L_t^\star\}_{t\geq 0}$ is a **nonnegative supermartingale** under every $Q \in \mathcal{P}_0$.



Choosing the rejection threshold

Ville's inequality. If $\{M_t\}_{t\geq 0}$ is a nonnegative supermartingale with $M_0 = 1$, then for every $B > 0$,

$$\mathbb{P}\left(\sup_{t\geq 0} M_t \geq B\right) \leq \frac{1}{B}.$$

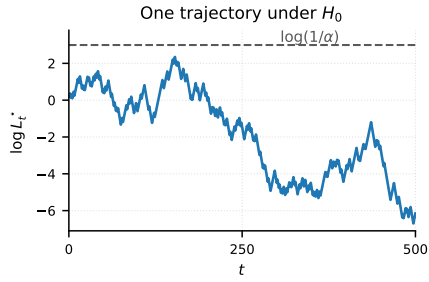
Apply this to the oracle process. Since $\{L_t^\star\}$ is a nonnegative supermartingale under every $Q \in \mathcal{P}_0$,

$$\sup_{Q\in\mathcal{P}_0} \mathbb{P}_Q\left(\sup_{t\geq 0} L_t^\star \geq B\right) \leq \frac{1}{B}.$$

Choose the threshold. Setting $B_\alpha = 1/\alpha$, define

$$\tau_\alpha^\star = \inf\{t \geq 1 : L_t^\star \geq 1/\alpha\}.$$

This immediately establishes the level-α property of $\tau_\alpha^\star$.



Power-one property and Log-optimality

Growth under P. Assuming integrable log-likelihood ratios, the strong law of large numbers gives

$$\frac{\log L_t^\star}{t} \xrightarrow{\text{a.s.}} \mathsf{D}(P\|Q^\star) = \mathsf{D}_{\inf}(P\|\mathcal{P}_0) > 0.$$

Power one. Since $\log L_t^\star \to \infty$ almost surely under P,

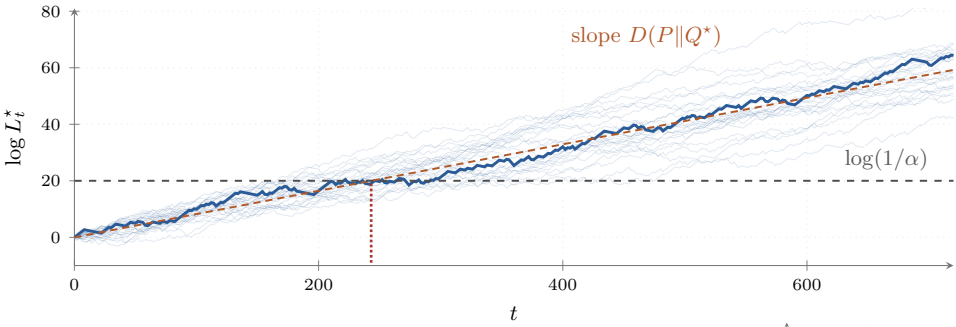
$$\mathbb{P}_P(\tau_\alpha^\star < \infty) = \mathbb{P}_P\left(\exists t \geq 1 : \frac{1}{t}\log L_t^\star \geq \frac{1}{t}\log(1/\alpha)\right) = 1.$$

Log-optimality. For any nonnegative $\mathcal{P}_0$-supermartingale $\{M_t\}_{t\geq 0}$ with $M_0 = 1$,

$$\mathbb{E}_P[\log M_t] \leq tD(P\|\mathcal{P}_0) = \mathbb{E}_P[\log L_t^\star].$$

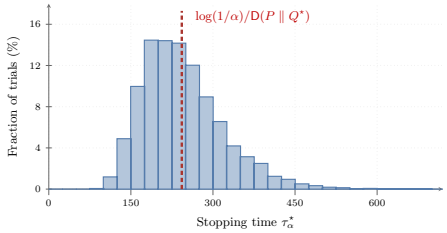
Thus, the oracle achieves the largest possible **expected logarithmic growth** under P.

How quickly does the oracle stop?



Under P, the log-evidence accumulates approximately linearly, with slope $\mathsf{D}(P\|Q^\star) = \mathsf{D}_{\inf}(P\|\mathcal{P}_0)$.

Typical crossing time: $\tau_\alpha^\star \approx \dfrac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P\|\mathcal{P}_0)}$



Expected stopping time of the oracle

1. Wald's identity. Since $\log L_t^\star = \sum_{i=1}^t \log \frac{dP}{dQ^\star}(X_i)$, we get

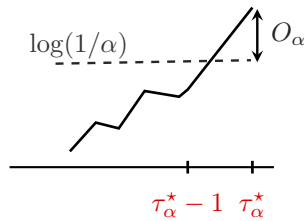
$$\mathbb{E}_P[\log L_{\tau_\alpha^\star}^\star] = \mathsf{D}(P \parallel \mathcal{P}_0)\, \mathbb{E}_P[\tau_\alpha^\star].$$

2. Overshoot control.

Define $O_\alpha := \log L_{\tau_\alpha^\star}^\star - \log(1/\alpha) \geq 0$

Under mild moment assumptions, Lorden's inequality gives

$$\mathbb{E}_P[O_\alpha] \leq C_P < \infty.$$



3. Conclusion. Combining the two results,

$$\mathbb{E}_P[\tau_\alpha^\star] = \frac{\log(1/\alpha) + O(1)}{\mathsf{D}_{\inf}(P\|\mathcal{P}_0)} \quad \Longrightarrow \quad \lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha^\star]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}$$

Lorden. *"On Excess Over the Boundary"* (1970)

Summary of the oracle test

1. Validity. The oracle likelihood-ratio process yields a level-α test, with power one under P:

$$\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha^\star < \infty) \le \alpha, \qquad \mathbb{P}_P(\tau_\alpha^\star < \infty) = 1, \quad \text{for all} \quad P \notin \mathcal{P}_0.$$

2. Optimality. Under mild assumptions, $\tau_\alpha^\star$ achieves the lower bound:

$$\mathbb{E}_P[\tau_\alpha^\star] = \frac{\log(1/\alpha) + O(1)}{\mathsf{D}_{\inf}(P \| \mathcal{P}_0)}.$$

3. The remaining challenge. $\tau_\alpha^\star$ requires knowing P and $Q^\star(P)$.

How do we attain the same performance without knowing P?

Constructive anytime-valid tests for finite alphabets

Finite alphabets: The oracle is unknown

$\{X_t : t \geq 1\} \overset{\text{i.i.d.}}{\sim} P$ lying in $\mathcal{X} = \{1, \ldots, k\}$.

Decide between $H_0 : P \in \mathcal{P}_0$ versus $H_1 : P \notin \mathcal{P}_0$

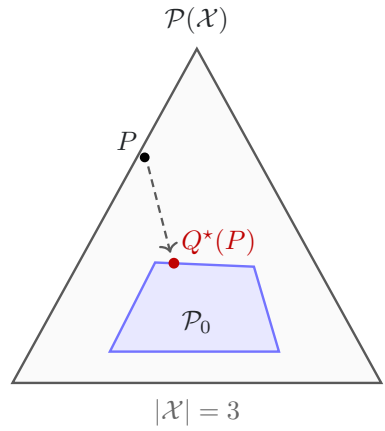
Recall the oracle process:

$$L_t^\star(P) = \prod_{i=1}^{t} \frac{P(X_i)}{Q^\star(X_i)},$$

$$Q^\star \equiv Q^\star(P) \in \arg\min_{Q \in \mathcal{P}_0} \mathsf{D}(P\|Q).$$

Oracle test: $\tau_\alpha^\star = \inf\{t \geq 1 : L_t^\star \geq 1/\alpha\}$

The difficulty. Both P and the associated RIPr $Q^\star(P)$ are unknown.



Two steps toward a data-driven test

Recall the oracle evidence process $\{L_t^\star : t \geq 1\}$, with $L_t^\star = \prod_{i=1}^t \frac{P(X_i)}{Q^\star(X_i)}$.

1. Simple null

$$\mathcal{P}_0 = \{Q_0\} \quad \Longrightarrow \quad Q^\star(P) = Q_0.$$

The denominator is known. We only need to **learn the alternative** P from past observations.

2. Composite null

$$Q^\star(P) \in \arg\min_{Q \in \mathcal{P}_0} D(P\|Q).$$

Both P and $Q^\star(P)$ are unknown. We must ensure **validity for every** $Q \in \mathcal{P}_0$.

Our route: first universal prediction for a simple null; then a construction that handles composite nulls.

Simple null and composite alternative: Validity

Assume $\mathcal{P}_0 = \{Q_0\}$, where $Q_0(x) > 0$ for every $x \in \mathcal{X}$.

$\{\widehat{P}_t : t \geq 1\}$: *any* sequence of predictable probability distributions

Construction. Define

$$M_0 = 1, \qquad M_t = M_{t-1}\frac{\widehat{P}_t(X_t)}{Q_0(X_t)}, \quad \text{and} \quad \tau_\alpha = \inf\{t \geq 1 : M_t \geq 1/\alpha\}.$$

Under Q_0,

$$\begin{aligned}\mathbb{E}_{Q_0}[M_t \mid \mathcal{F}_{t-1}] &= M_{t-1}\sum_{x\in\mathcal{X}} Q_0(x)\frac{\widehat{P}_t(x)}{Q_0(x)} \\ &= M_{t-1}\sum_{x\in\mathcal{X}} \widehat{P}_t(x) = M_{t-1}.\end{aligned}$$

Takeaway. Any predictable $\{\widehat{P}_t : t \geq 1\}$ ensures level-α property of τ_α under Q_0.

But which predictor should we choose?

Universal prediction

When $P \neq Q_0$, we need **good predictions** for rapid evidence growth

A Bayesian mixture. Let $k = |\mathcal{X}|$ and $\pi = \text{Dirichlet}(1/2, \ldots, 1/2)$. Define

$$\widetilde{P}(x^t) = \int_{\Delta_{k-1}} \prod_{i=1}^{t} q(x_i)\, \pi(dq).$$

The resulting sequential KT predictor:

$$\widehat{P}_t(x) \equiv \widehat{P}_t(x \mid X^{t-1}) = \frac{\widetilde{P}(X^{t-1}x)}{\widetilde{P}(X^{t-1})} = \frac{N_{t-1}(x) + \frac{1}{2}}{t - 1 + \frac{k}{2}} = \quad \text{“add 1/2” estimator.}$$

For Bernoulli observations in A/B testing example, this gives

$$\widehat{p}_t = \frac{S_{t-1} + 1/2}{t}.$$

The price of learning the alternative

Evidence decomposition. For the simple null $\mathcal{P}_0 = \{Q_0\}$, and true distribution P,

$$\log M_t = \log \frac{\widetilde{P}(X^t)}{Q_0^t(X^t)} = \log \frac{P^t(X^t)}{Q_0^t(X^t)} \frac{\widetilde{P}(X^t)}{P^t(X^t)} = \underbrace{\log \frac{P^t(X^t)}{Q_0^t(X^t)}}_{\text{Oracle evidence } \log L_t^\star} - \underbrace{\log \frac{P^t(X^t)}{\widetilde{P}(X^t)}}_{\text{Cost of not knowing } P}.$$

Universal coding regret. The KT predictor guarantees, for every x^t,

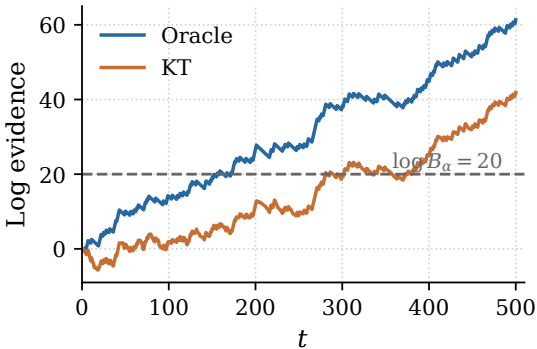
$$R_t(P) := \log \frac{P^t(X^t)}{\widetilde{P}(X^t)} \leq \frac{k-1}{2} \log(t+1) + C_k.$$

Hence, we have

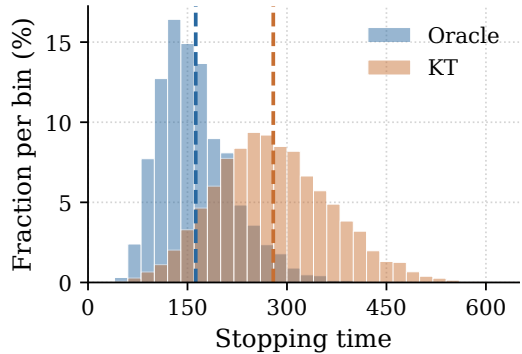
$$\log M_t \geq \underbrace{\log L_t^\star}_{\text{grows linearly}} - O(\log t).$$

Universal prediction versus the oracle

$k = 10$, $\mathsf{D}(P\|Q_0) = 0.125$ and $\log(1/\alpha) = 20$.



KT trails oracle on the same observations.



Expected stopping times: 162 (oracle), 279 (KT).

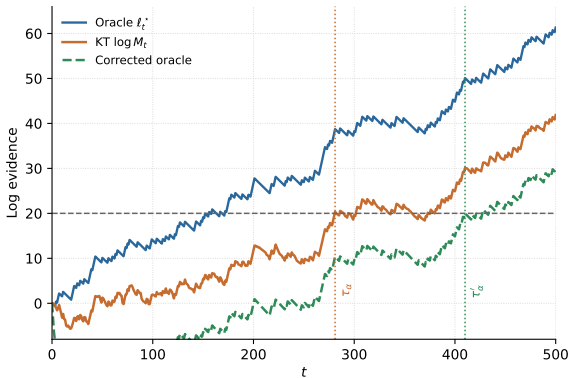
From regret to stopping time

$\log M_t \geq \log L_t^\star - r_t$ where $r_t = O(\log t)$

Recall the stopping time definition

$$\begin{aligned}\tau_\alpha &= \inf\{t \geq 1 : \log M_t \geq \log(1/\alpha)\} \\ &\geq \inf\{t \geq 1 : \log L_t^\star - r_t \geq \log(1/\alpha)\} \\ &:= \tau'_\alpha.\end{aligned}$$

τ'_α: first time the oracle process hits the regret-corrected threshold.

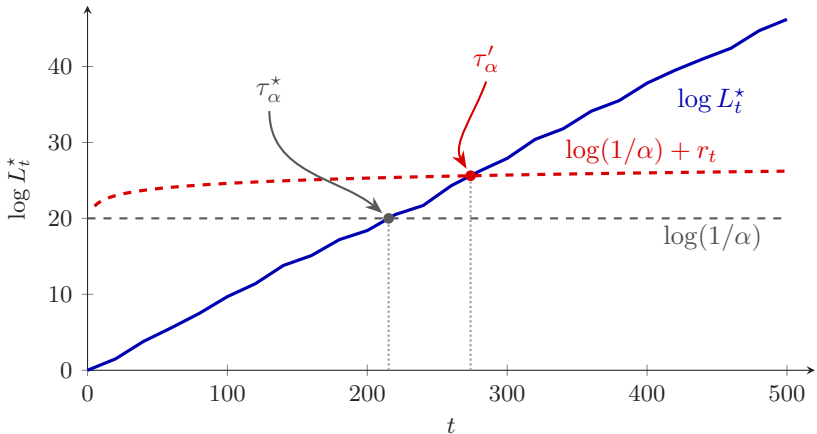


Combining with the lower bound

$$\underbrace{\frac{\log(1/\alpha)}{\mathsf{D}(P \,\|\, Q_0)}}_{\text{Converse}} \leq \mathbb{E}_P[\tau_\alpha] \leq \underbrace{\mathbb{E}_P[\tau'_\alpha]}_{\text{Next slide}}.$$

Simple null: first-order optimality

$$\tau'_\alpha = \inf\left\{t \geq 1 : \log L^\star_t \geq \log(1/\alpha) + r_t\right\}, \qquad r_t = O(\log t).$$



Simple null: first-order optimality

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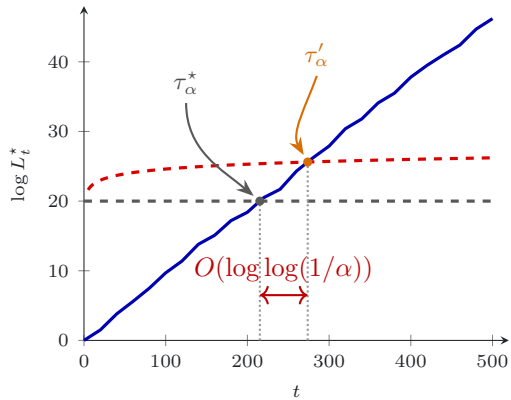
Expected stopping-time bound

$$\mathbb{E}_P[\tau'_\alpha] \leq \frac{\log(1/\alpha) + O(\log\log(1/\alpha))}{\mathsf{D}(P \parallel Q_0)}.$$

From the previous slide,

$$\frac{\log(1/\alpha)}{\mathsf{D}(P \parallel Q_0)} \leq \mathbb{E}_P[\tau_\alpha] \leq \mathbb{E}_P[\tau'_\alpha].$$

$$\lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}(P \parallel Q_0)}.$$

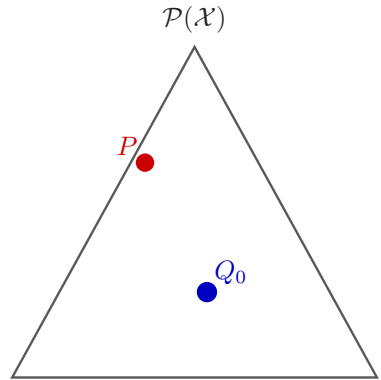


From simple to composite nulls

Simple null: $\mathcal{P}_0 = \{Q_0\}$.

$$M_t^{Q_0} = \frac{\widetilde{P}(X^t)}{Q_0^t(X^t)}, \qquad Q_0^t(X^t) = \prod_{i=1}^{t} Q_0(X_i).$$

One null distribution $\Longrightarrow$ one martingale.



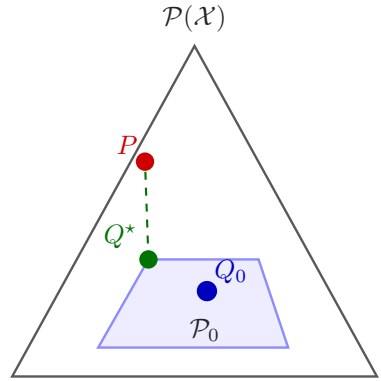
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Composite null: Ideal choice of denominator is

$$Q^\star \equiv Q^\star(P) \in \arg\min_{Q \in \mathcal{P}_0} Q(X^t).$$



Idea: take the infimum over all null distributions.

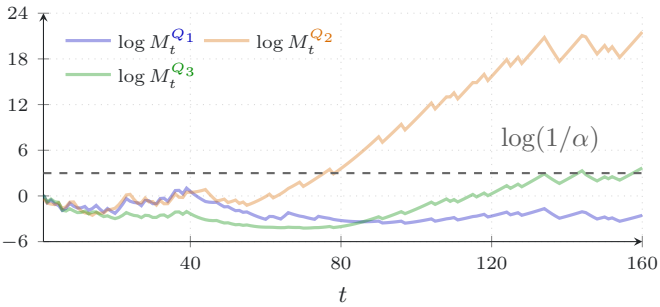
$$E_t := \inf_{Q \in \mathcal{P}_0} M_t^Q = \frac{\widetilde{P}(X^t)}{\sup_{Q \in \mathcal{P}_0} Q^t(X^t)} = \frac{\text{Mixture of likelihoods}}{\text{Maximum null likelihood}}.$$

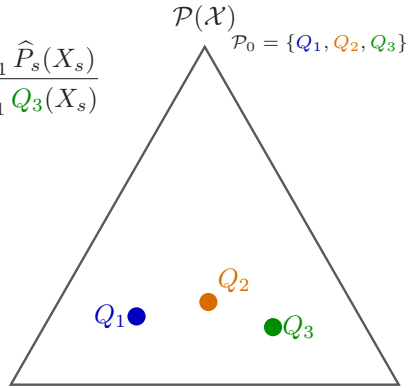
Behavior of E_t under the null

$$M_t^{Q_1} = \frac{\prod_{s=1}^t \widehat{P}_s(X_s)}{\prod_{s=1}^t Q_1(X_s)}, \quad M_t^{Q_2} = \frac{\prod_{s=1}^t \widehat{P}_s(X_s)}{\prod_{s=1}^t Q_2(X_s)}, \quad M_t^{Q_3} = \frac{\prod_{s=1}^t \widehat{P}_s(X_s)}{\prod_{s=1}^t Q_3(X_s)}$$

$$E_t := \min\{M_t^{Q_1}, M_t^{Q_2}, M_t^{Q_3}\}.$$

Suppose H_0 holds, and the true distribution is Q_1.





$E_t \leq M_t^{Q_1}$ for all t.

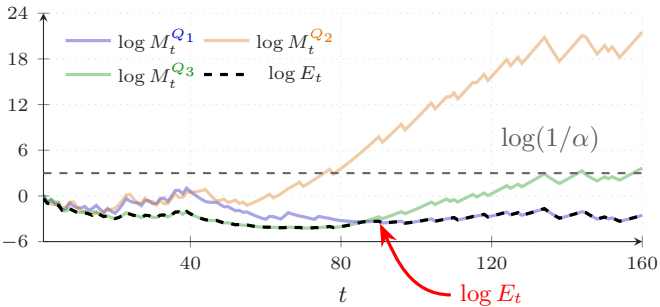
If Q_1 is true, then $\{M_t^{Q_1} : t \geq 0\}$ is a martingale.

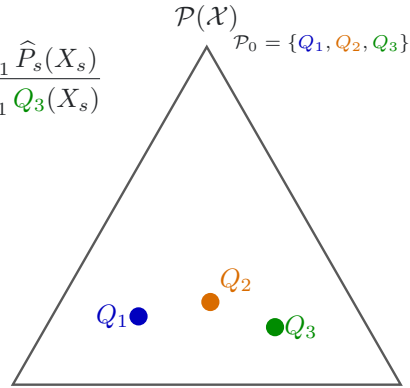
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$$E_t := \min\{M_t^{Q_1}, M_t^{Q_2}, M_t^{Q_3}\}.$$

Suppose H_0 holds, and the true distribution is Q_1.



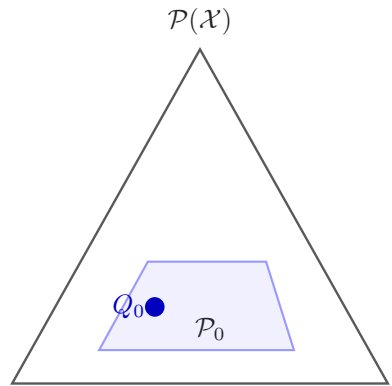


$E_t \le M_t^{Q_1}$ for all t.

If Q_1 is true, then $\{M_t^{Q_1} : t \ge 0\}$ is a martingale.

Uniform Level-α under general composite $\mathcal{P}_0$

$$M_t^Q := \frac{\prod_{i=1}^t \widehat{P}_i(X_i)}{\prod_{i=1}^t Q(X_i)}, \qquad E_t := \inf_{Q \in \mathcal{P}_0} M_t^Q.$$



If $X_1, X_2, \ldots \sim Q_0$ for $Q_0 \in \mathcal{P}_0$, we have

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q \leq M_t^{Q_0}.$$

$\{M_t^{Q_0}\}_{t \geq 0}$ is a nonnegative martingale under Q_0.

$\{E_t\}_{t \geq 0}$ is dominated by a Q_0-martingale $\forall Q_0 \in \mathcal{P}_0$.

This implies **uniform validity** under $\mathcal{P}_0$:

$$\sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\exists t \geq 1 : E_t \geq \frac{1}{\alpha}\right) \leq \sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\exists t \geq 1 : M_t^{Q_0} \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Two prices of ignorance when $P \notin \mathcal{P}_0$

Full oracle
Knows P and $Q^\star(P)$.

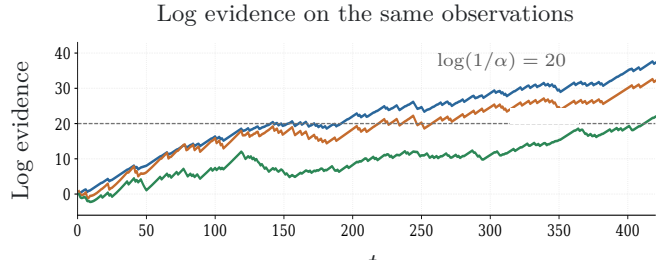
$$L_t^\star = \frac{P^t(X^t)}{Q^{\star\, t}(X^t)}.$$

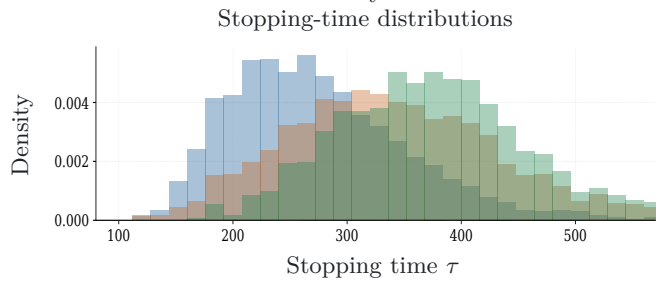
Semi-oracle benchmark
Knows $Q^\star(P)$; learns P by KT.

$$M_t^{Q^\star} = \frac{\widetilde{P}(X^t)}{Q^{\star\, t}(X^t)}.$$

Universal inference
Learns P and protects against every null.

$$E_t = \frac{\widetilde{P}(X^t)}{\sup\limits_{Q \in \mathcal{P}_0} Q^t(X^t)}.$$





Behavior of $\{E_t\}$ when $P \notin \mathcal{P}_0$

Empirical distribution: $\widehat{\mu}_t = \frac{1}{t}\sum_{i=1}^{t} \delta_{X_i}$

We can show that

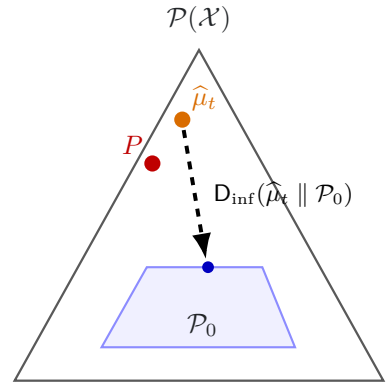
$$\log E_t = \underbrace{t\,\mathsf{D}_{\inf}(\widehat{\mu}_t \parallel \mathcal{P}_0)}_{\text{Empirical information}} - \underbrace{\log \frac{\widehat{\mu}_t^t(X^t)}{\widetilde{P}(X^t)}}_{\text{KT regret}}.$$

KT regret bound implies that

$$\log \frac{\widehat{\mu}_t^t(X^t)}{\widetilde{P}(X^t)} \le \frac{k-1}{2}\log(t+1) + C_k.$$

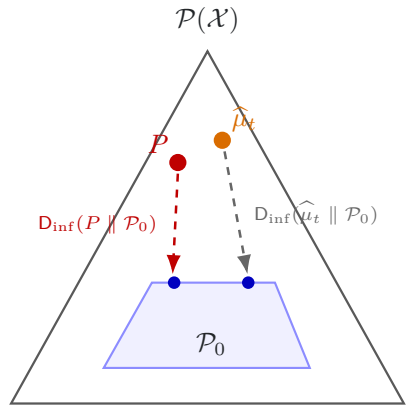
Hence,

$$\log E_t \ge t\,\mathsf{D}_{\inf}(\widehat{\mu}_t \parallel \mathcal{P}_0) - O(\log t).$$



Composite nulls: first-order optimality

$$\log E_t \geq t\, \mathsf{D}_{\inf}(\widehat{\mu}_t \parallel \mathcal{P}_0) - O(\log t).$$



Composite nulls: first-order optimality

$$\log E_t \geq t\, \mathsf{D}_{\inf}(\widehat{\mu}_t \parallel \mathcal{P}_0) - O(\log t).$$

Under P, we have

$$\mathsf{D}_{\inf}(\widehat{\mu}_t \parallel \mathcal{P}_0) \longrightarrow \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0).$$

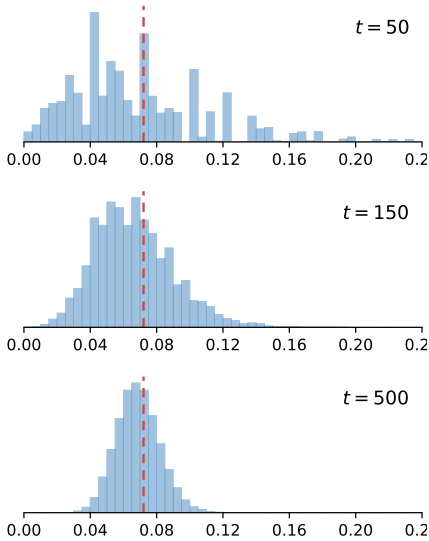
Lower-tail deviations are exponentially rare.

We can use this fact to show

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha) + o(\log(1/\alpha))}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}.$$

Together with the converse, this implies

$$\lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}$$



Recap of Part I: Anytime-Valid Testing

Problem. Given an i.i.d. stream of $\mathcal{X}$-valued observations, decide between

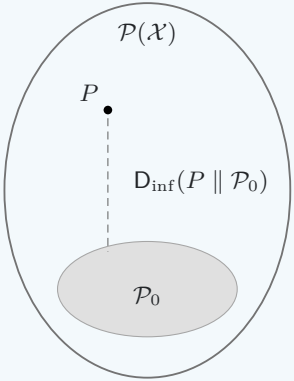
$$H_0 : P_X \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P_X \in \mathcal{P}_1.$$

Goal. Construct a level-α, power-one test τ_α satisfying

$$\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha < \infty) \le \alpha, \qquad \inf_{P \in \mathcal{P}_1} \mathbb{P}_P(\tau_\alpha < \infty) = 1.$$

Fundamental limit. Every such test satisfies

$$\mathbb{E}_P[\tau_\alpha] \ge \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}, \qquad P \in \mathcal{P}_1.$$



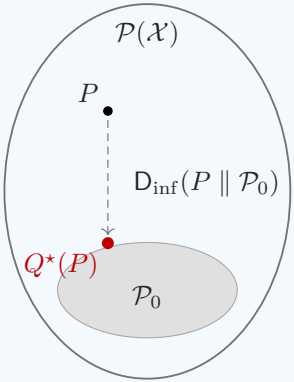
$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) = \inf_{Q \in \mathcal{P}_0} \mathsf{D}(P \| Q)$

Recap of Part I: The oracle construction

General approach. Construct a process $\{M_t : t \geq 1\}$ that

- is unlikely to grow under H_0,
- grows rapidly under H_1.

$$\tau_\alpha = \inf\{t \geq 1 : M_t \geq 1/\alpha\}.$$



Oracle construction. If the alternative P is known,

$$L_t^\star = \prod_{i=1}^{t} \frac{dP}{dQ^\star(P)}(X_i), \quad \text{where} \quad Q^\star(P) \in \arg\min_{Q \in \mathcal{P}_0} \mathsf{D}(P \parallel Q),$$

Validity and optimality. Under suitable assumptions on $\mathcal{P}_0$,

- $L_t^\star$ is a nonnegative supermartingale under every $Q \in \mathcal{P}_0$.
- The resulting test is first-order optimal: $\mathbb{E}_P[\tau_\alpha^\star] \sim \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \| \mathcal{P}_0)}$.

Recap of Part I: Finite $\mathcal{X}$ + Simple Null

Simple null: $\mathcal{P}_0 = \{Q_0\}$, hence $Q^\star(P) = Q_0$ for all P

Idea 1: Universal prediction. Replace P by the KT predictor:

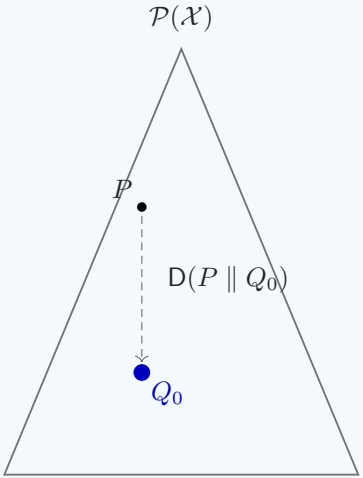
$$M_t = \frac{\widetilde{P}(X^t)}{Q_0^t(X^t)}, \qquad \widetilde{P}(X^t) = \prod_{i=1}^{t} \widehat{P}_i(X_i).$$

Cost of learning: $O(\log t)$ regret.

$$\log M_t \geq \log L_t^\star - O(\log t).$$

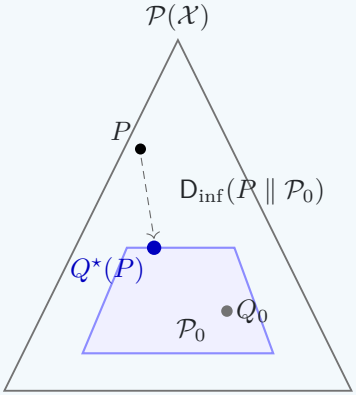
Result: Validity and first-order optimality.

$$\lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}(P \parallel Q_0)}.$$



Recap of Part I: Finite $\mathcal{X}$ + Composite Null

Composite null: $\mathcal{P}_0 \subseteq \mathcal{P}(\mathcal{X})$. The alternative P is unknown.



Idea 2: Pointwise domination. Protect against every $Q \in \mathcal{P}_0$:

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q = \frac{\widetilde{P}(X^t)}{\sup_{Q \in \mathcal{P}_0} Q^t(X^t)}.$$

For every $Q_0 \in \mathcal{P}_0$, $E_t \leq M_t^{Q_0}$

Result: Uniform validity and first-order optimality.

$$\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q\left(\tau_\alpha < \infty\right) \leq \alpha, \qquad \lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)}.$$

Beyond finite alphabets: dominated models

Parametric models: $\{P_\theta : \theta \in \Theta \subseteq \mathbb{R}^d\}$, with densities p_θ.

$$\mathcal{P}_0 = \{P_\theta : \theta \in \Theta_0\}, \qquad \Theta_0 \subseteq \Theta.$$

E.g., Gaussian location, Poisson, and exponential families.

Same two ideas: Universal prediction + maximum null likelihood.

$$E_t = \frac{\displaystyle\int_\Theta \prod_{i=1}^{t} p_\theta(X_i)\, \pi(d\theta)}{\displaystyle\sup_{\eta \in \Theta_0} \prod_{i=1}^{t} p_\eta(X_i)}.$$

Under suitable regularity conditions:

$$\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha < \infty) \le \alpha, \qquad \lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}.$$

Looking ahead to Part II

Part I: Universal prediction + null MLE

$$\Longrightarrow \quad \text{Anytime validity and first-order optimality.}$$

But what if likelihood models are unavailable?

Example: Bounded-mean testing: For some $m \in (0,1)$, consider

$$X_t \in [0,1], \qquad H_0 : \mathbb{E}_P[X] \leq m \quad \text{vs.} \quad H_1 : \mathbb{E}_P[X] > m.$$

The null contains **arbitrary distributions** on $[0,1]$ with mean at most m, with no common dominating measure.

Can we construct first-order-optimal anytime-valid tests without likelihoods?

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